

HOUSE PRICE PREDICTION

Educational Practice Project

Project Title: House Price Prediction Using Machine Learning

Group: BDA-2513

Course: Educational Practice

University: Astana IT University

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Project Track: Machine Learning Project

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Team Members

Student	Contribution
Makhanbet Alikhan	Dataset collection and Exploratory Data Analysis (EDA)
Matmusaev Alikhan	Data preprocessing and feature engineering
Aitzhanov Alikhan	Machine learning model training and evaluation
Adilbek Nurdaulet	Report writing, README preparation, screenshots collection and demo video

1. Project Overview

The goal of this project is to build a machine learning model that can predict house sale prices based on various house characteristics. The project demonstrates a complete machine learning workflow including data exploration, preprocessing, feature engineering, model training, prediction, and evaluation.

The project was completed as part of the Educational Practice course and follows industry-style development stages where different team members are responsible for different phases of the project.

2. Problem Statement

House prices depend on many factors such as house size, quality, location, number of rooms, garage size, and year of construction.

Manually estimating house prices can be difficult because many variables influence the final value.

The objective of this project is to develop a machine learning model capable of predicting the sale price of a house using historical housing data.

Target variable:

SalePrice

3. Methodology

The project was divided into three main stages:

Stage 1: Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset and identify important patterns.

Activities included:

- Dataset inspection
- Missing value analysis
- Distribution analysis
- Correlation analysis
- Feature importance exploration
- Outlier detection

Several visualizations were created to better understand the housing data.

Stage 2: Data Preprocessing

The preprocessing stage prepared the dataset for machine learning.

Tasks performed:

- Handling missing values
- Encoding categorical variables
- Removing unnecessary information
- Creating new features
- Preparing a clean dataset for model training

The final processed dataset was saved as:

db_preprocessed.csv

Stage 3: Machine Learning

The machine learning stage focused on building a predictive model.

Steps:

1. Load processed dataset
 2. Separate features and target variable
 3. Split data into training and validation sets
 4. Train model
 5. Generate predictions
 6. Evaluate performance
 7. Analyze feature importance
-

4. Tools and Technologies Used

Programming Language:

- Python

Libraries:

- Pandas
- NumPy
- Scikit-learn

Version Control:

- GitHub
-

5. Model Description

The project uses:

RandomForestRegressor

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Advantages:

- Good performance on tabular data
- Handles large numbers of features
- Reduces overfitting
- Easy to interpret using feature importance

Model parameters:

```
RandomForestRegressor(  
    n_estimators=100,  
    random_state=42,  
    n_jobs=-1  
)
```

6. Implementation Explanation

The implementation follows the standard machine learning pipeline.

Data Loading

The processed dataset is loaded into a pandas DataFrame.

Feature Selection

Target variable:

SalePrice

Removed column:

Id

The Id column was removed because it does not contain useful information for prediction.

Train-Test Split

The dataset was split into:

- 80% training data
- 20% validation data

This allows the model to be evaluated on unseen examples.

Model Training

The Random Forest model was trained using the training dataset.

Prediction

Predictions were generated on the validation dataset.

Evaluation

The model was evaluated using:

- MAE
 - RMSE
 - R^2 Score
-

7. Implementation Details

The machine learning model was implemented using Python and Scikit-learn.

Data Loading

The preprocessed dataset (db_preprocessed.csv) was loaded into a Pandas DataFrame. This dataset was already cleaned and prepared during the preprocessing stage.

Feature Selection

The target variable selected for prediction was:

SalePrice

The **Id** column was removed because it is only an identifier and does not provide useful information for house price prediction.

Train-Test Split

The dataset was divided into training and validation sets using the `train_test_split()` function.

- Training data: 80%
- Validation data: 20%

This approach allows the model to be evaluated on unseen data and helps measure its generalization performance.

Model Training

A **RandomForestRegressor** model was used for training.

The model parameters were:

- `n_estimators = 100`
- `random_state = 42`
- `n_jobs = -1`

The model was trained using the training dataset and learned relationships between house characteristics and sale prices.

Prediction Generation

After training, the model generated predictions for the validation dataset.

The predicted values were compared with actual house prices to evaluate model performance.

Model Evaluation

The model was evaluated using three regression metrics:

- **MAE (Mean Absolute Error)** – measures the average prediction error.
- **RMSE (Root Mean Squared Error)** – gives higher penalty to large errors.
- **R² Score** – measures how well the model explains the variance of house prices.

Final results:

- MAE = 16642.90
- RMSE = 24667.54
- R² Score = 0.8898

Feature Importance Analysis

Random Forest provides feature importance values that show which variables have the greatest influence on predictions.

The most important features identified by the model were:

1. TotalSF
2. OverallQual
3. GrLivArea
4. HouseAge
5. BsmtFinSF1

These features were found to have the strongest impact on house sale prices.

8. Screenshots

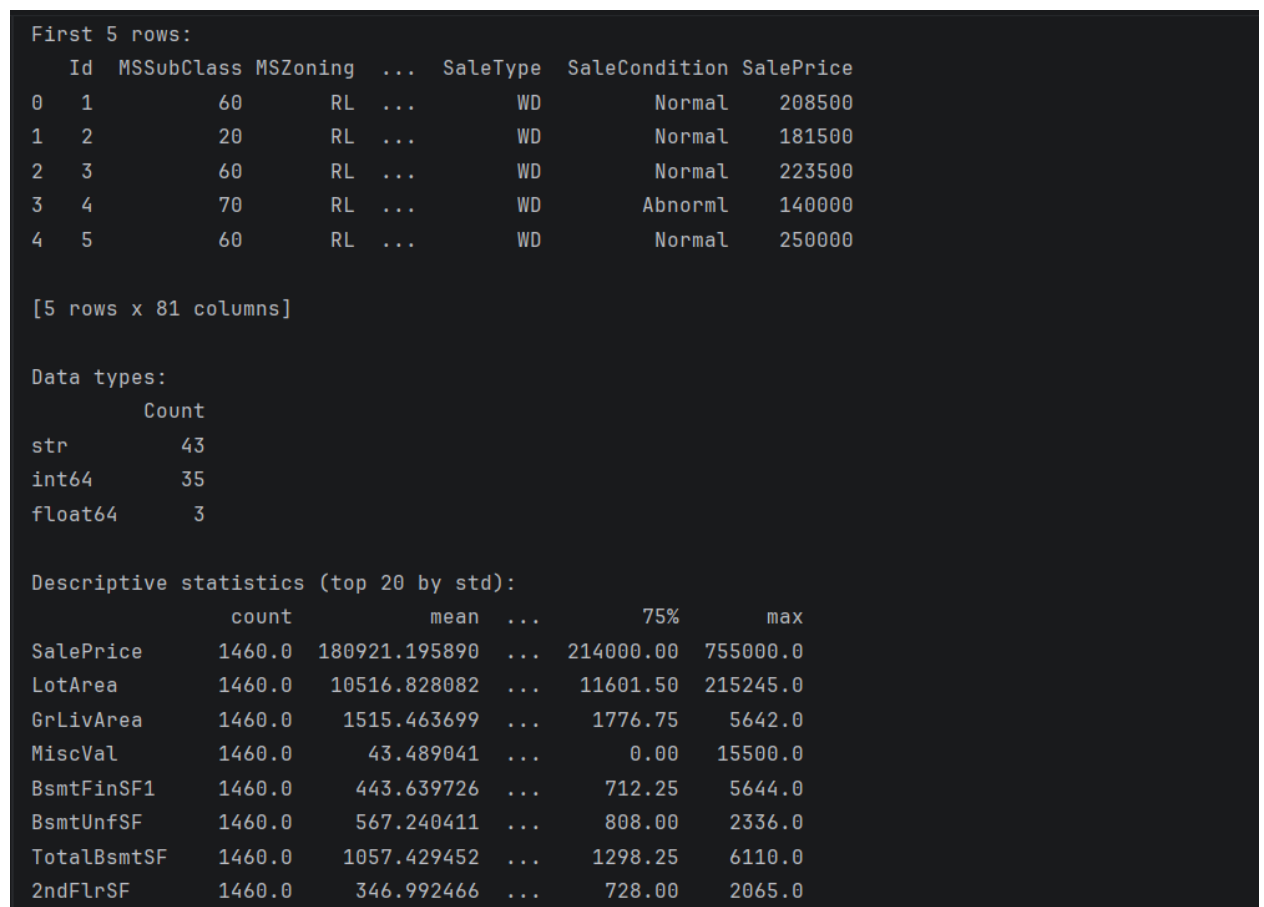


Figure 1. Dataset Information

DB: 19 columns with missing values

	Missing	Missing_%
PoolQC	1453	99.52
MiscFeature	1406	96.30
Alley	1369	93.77
Fence	1179	80.75
MasVnrType	872	59.73
FireplaceQu	690	47.26
LotFrontage	259	17.74
GarageType	81	5.55
GarageYrBlt	81	5.55
GarageFinish	81	5.55
GarageQual	81	5.55
GarageCond	81	5.55
BsmtExposure	38	2.60
BsmtFinType2	38	2.60
BsmtQual	37	2.53
BsmtCond	37	2.53
BsmtFinType1	37	2.53
MasVnrArea	8	0.55
Electrical	1	0.07

Figure 2. Missing Values Analysis

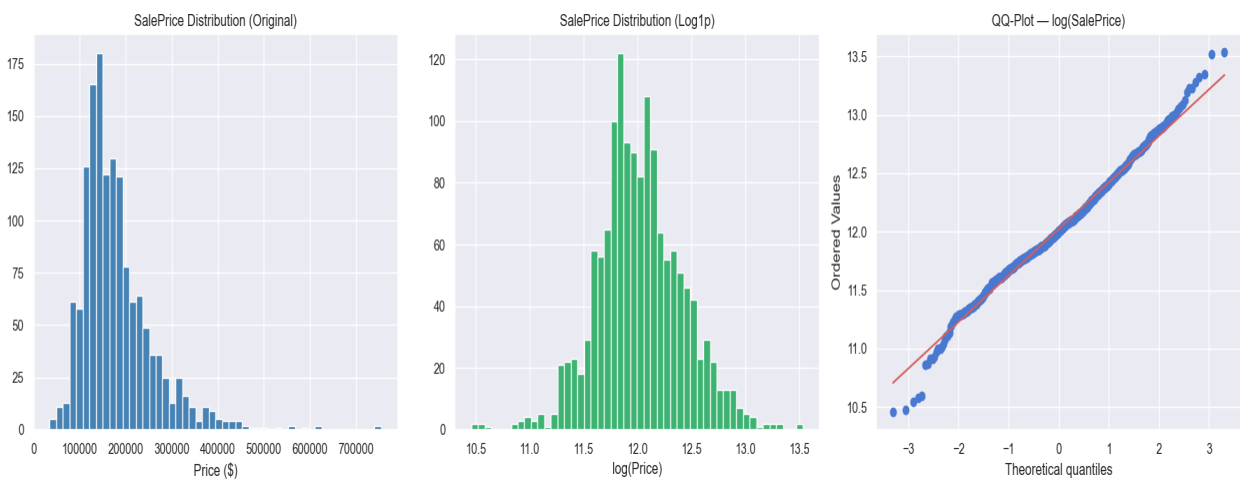


Figure 3. Sale Price Distribution

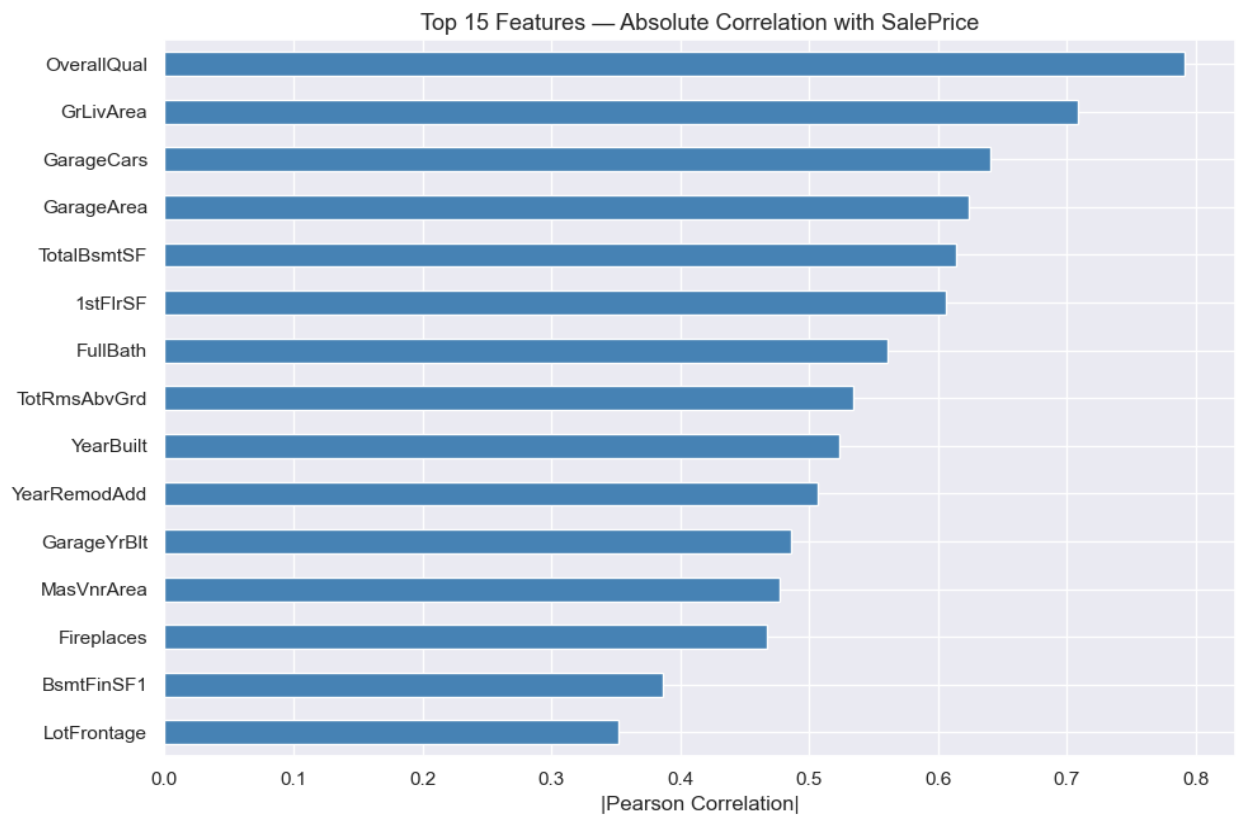


Figure 4. Top Correlation

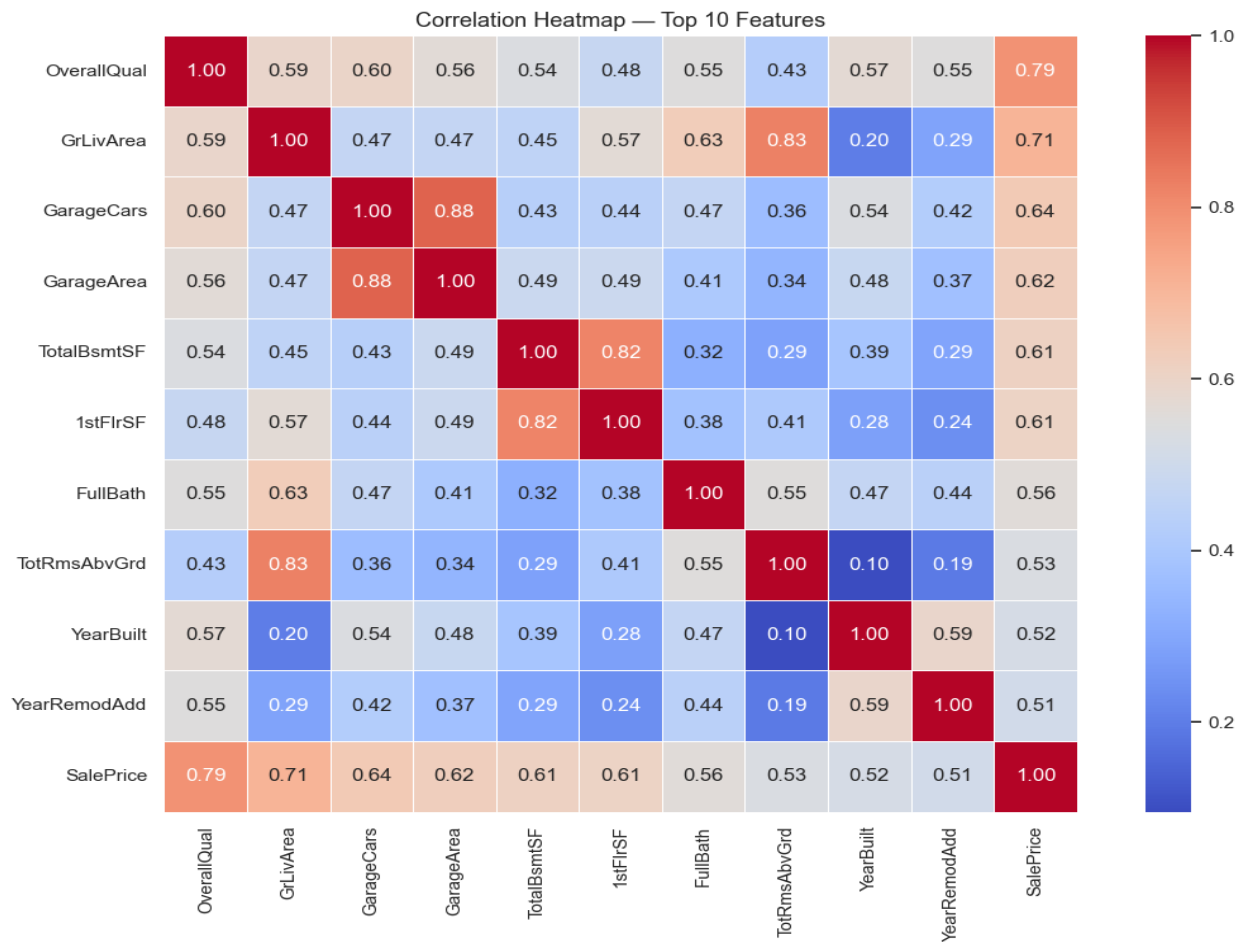


Figure 5. Correlation Heatmap

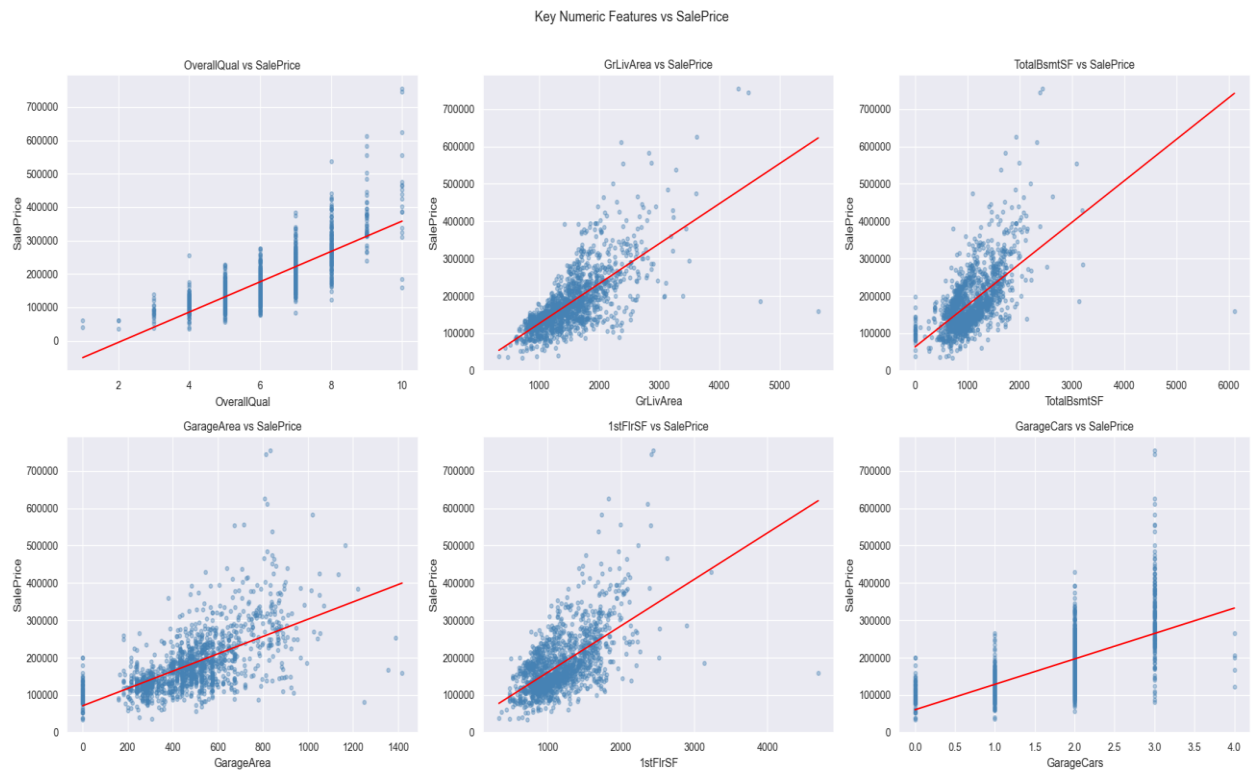


Figure 6. Important Features

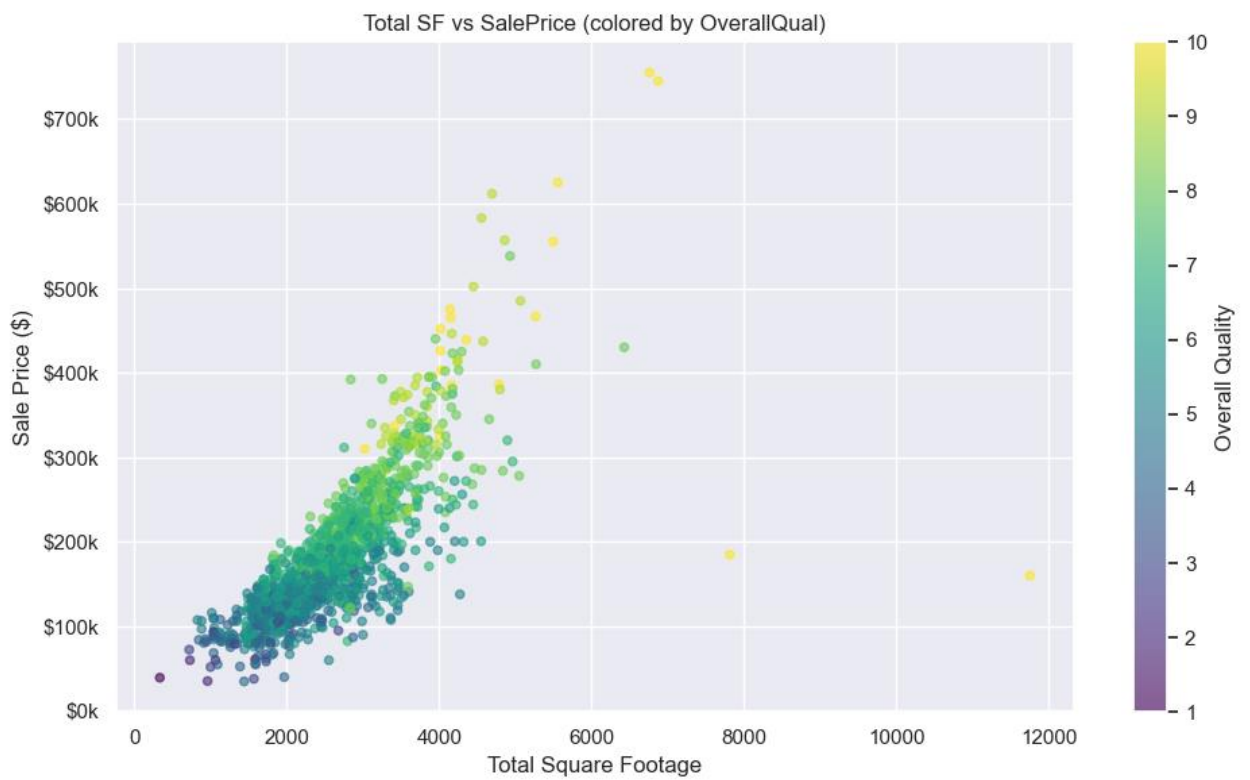


Figure 7. TotalSF Analysis

```

# Remove outliers
db = db[db["GrLivArea"] < 4500]
print(f"\nShape after removing outliers: {db.shape}")

# Fill missing values - numerical
db["LotFrontage"] = db["LotFrontage"].fillna(db["LotFrontage"].median())
db["MasVnrArea"] = db["MasVnrArea"].fillna(0)

# Fill missing values - basement columns
bsmt_cols = ["BsmtQual", "BsmtCond", "BsmtExposure", "BsmtFinType1", "BsmtFinType2"]
for col in bsmt_cols:
    db[col] = db[col].fillna("None")

# Fill missing values - garage columns
garage_cols = ["GarageType", "GarageFinish", "GarageQual", "GarageCond"]
for col in garage_cols:
    db[col] = db[col].fillna("None")

# Fill remaining categorical and numerical columns
db["MasVnrType"] = db["MasVnrType"].fillna("None")
db["Electrical"] = db["Electrical"].fillna(db["Electrical"].mode()[0])

for col in db.select_dtypes(include="object").columns:
    db[col] = db[col].fillna("None")

for col in db.select_dtypes(include="number").columns:
    db[col] = db[col].fillna(db[col].median())

print(f"Missing values left: {db.isnull().sum().sum()}")

```

Figure 8. Missing Value Processing

```
# Feature engineering
db["HouseAge"] = db["YrSold"] - db["YearBuilt"]
db["RemodAge"] = db["YrSold"] - db["YearRemodAdd"]
db["TotalBathrooms"] = (
    db["FullBath"] + 0.5 * db["HalfBath"] + db["BsmtFullBath"] + 0.5 * db["BsmtHalfBath"]
)
db["HasGarage"] = (db["GarageArea"] > 0).astype(int)
db["HasBasement"] = (db["TotalBsmtSF"] > 0).astype(int)
print("New features created: HouseAge, RemodAge, TotalBathrooms, HasGarage, HasBasement")

# Label encoding for categorical columns
le = LabelEncoder()
for col in db.select_dtypes(include="object").columns:
    db[col] = le.fit_transform(db[col].astype(str))
print("Label encoding completed")

print(f"Final dataset shape: {db.shape}")
print(db.head())
```

Figure 9. Feature Engineering and Encoding

```
New features created: HouseAge, RemodAge, TotalBathrooms, HasGarage, HasBasement
Label encoding completed
Final dataset shape: (1458, 87)
```

	Id	MSSubClass	MSZoning	...	TotalBathrooms	HasGarage	HasBasement
0	1	60	3	...	3.5	1	1
1	2	20	3	...	2.5	1	1
2	3	60	3	...	3.5	1	1
3	4	70	3	...	2.0	1	1
4	5	60	3	...	3.5	1	1

Figure 10. Feature Importance after prediction

9. Evaluation Results

The final machine learning model achieved the following results:

Metric	Value
MAE	16642.90
RMSE	24667.54
R ² Score	0.8898

These results indicate that the model performs well and explains approximately 89% of the variance in house prices.

10. Most Important Features

According to the Random Forest model, the most important features are:

1. TotalSF
2. OverallQual
3. GrLivArea
4. HouseAge
5. BsmtFinSF1

These variables significantly influence house sale prices.

The results are reasonable because larger and higher-quality houses generally have higher market values.

11. Difficulties and Solutions

Difficulty 1

The dataset contained missing values.

Solution:

Missing values were identified and replaced using appropriate preprocessing techniques.

Difficulty 2

The dataset included categorical features.

Solution:

Categorical variables were encoded into numerical representations.

Difficulty 3

Different team members worked on separate stages.

Solution:

The project was divided into EDA, preprocessing, and machine learning components, allowing efficient collaboration.

12. Conclusion

This project successfully implemented a complete machine learning workflow for house price prediction.

The team performed:

- Data exploration
- Data preprocessing
- Feature engineering
- Machine learning model training
- Prediction generation
- Model evaluation

The Random Forest model achieved strong performance with an R^2 score of approximately 0.89, demonstrating its effectiveness for predicting housing prices.

The project provided practical experience in machine learning, teamwork, data analysis, and software development workflows.

13. GitHub Repository and Demo Video

GitHub Repository

<https://github.com/aitu-educational-practice-2026/educational-practice-project-priora>

Demo Video

<https://drive.google.com/drive/folders/1X2LF5jxc8JMHARCsvvfyF2YlR5ThcoAc?usp=sharing>

Contribution Table

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